

Workforce Insights Dashboard for Employee Skill and Analytics

Milestone 1 Documentation

Organization: Infosys Springboard Internship

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1. Introduction

Modern organizations generate workforce data from multiple HR systems such as SAP SuccessFactors, Workday, Learning Management Systems (LMS), recruitment portals, and employee engagement platforms. These systems store employee information in different formats, making it difficult to perform centralized workforce analytics.

The **AI-Powered Workforce Analytics & Talent Intelligence Dashboard** aims to integrate workforce data into a centralized repository, enabling organizations to monitor employee performance, analyze workforce trends, predict employee attrition, identify skill gaps, and support data-driven HR decisions. The first milestone focuses on building the foundation of this platform by defining data requirements, designing the repository architecture, creating the database schema, planning secure integration mechanisms, and establishing ETL workflows.

2. Project Objectives

The objectives of Milestone 1 are:

- Study workforce data requirements.
- Define system analytics specifications.
- Design repository architecture.
- Create the database schema and ER design.
- Design ETL and data integration workflows.
- Develop secure connectivity mechanisms for SAP and Workday.
- Implement data cleansing and transformation workflows.
- Design data storage workflows.
- Prepare clean data for AI-powered analytics.
- Build the foundation for future dashboard implementation.

3. Team Members and Responsibilities

Team Member	Responsibility
Navya	Workforce Data Requirements
Tanisha Gehlot	Analytics Specifications
Gowshika T	Repository Architecture
Swetha Sneharaj	Database Schema & ER Diagram
Lakshmikanth Reddy	ETL & Data Integration
Kavya	SAP Connectivity
Nikhil	Workday Connectivity
Sanjay bhargav	Data Cleansing Workflow
Santhoshini	Data Transformation Workflow
G. Lakshmi Keertana	Data Storage Workflow, Final Documentation & Presentation

4. Workforce Data Requirements

The workforce analytics repository collects information from multiple HR domains.

Employee Information

- Employee ID

- Employee Name
- Gender
- Age
- Department
- Job Role
- Location
- Hire Date
- Employment Status
- Attendance

Payroll Data

- Salary
- Bonus
- Incentives
- Stock Options

Performance Data

- Performance Rating
- Manager Rating
- Self Rating
- Promotion History

Attrition Data

- Exit Date
- Exit Reason
- Attrition Status
- Years at Company

Learning Data

- Skills
- Training Hours
- Certifications
- Education

These data requirements support workforce planning, employee engagement analysis, and AI-based insights.

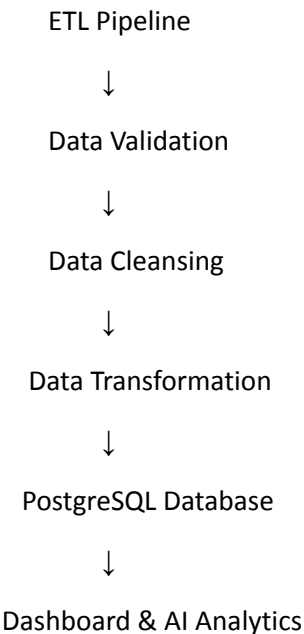
5. System Analytics Specifications

The dashboard will provide analytics through KPIs and reports.

Key Performance Indicators

- Total Employees
- Attrition Rate
- Promotion Rate
- Average Salary
- Attendance Percentage
- Job Satisfaction Score
- Workforce Health Score
- Training Completion
- High Attrition Employees

System work flow

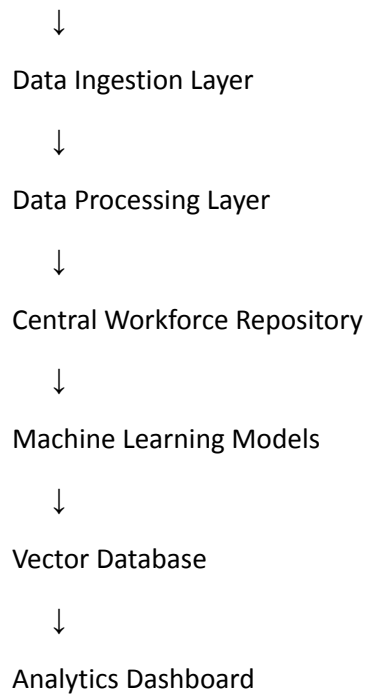


The dashboard includes workforce overview, employee analytics, performance, attendance, salary, promotion, attrition, and AI insights modules.

6. Repository Architecture

The repository architecture follows a layered approach:

Data Sources



The architecture supports ingestion, preprocessing, centralized storage, AI models, and dashboard visualization.

7. Database Schema Design

The database stores workforce information using multiple interconnected tables.

Major Tables

- Organization
- Department
- Employee
- Job Position
- Performance Review
- Performance Goal
- Attrition Record
- Engagement Survey
- Skills
- Learning Records
- Recruitment
- Candidate
- Data Source

- Data Ingestion Log
- Workforce Metrics
- System User

The schema uses primary keys, foreign keys, and relationships to maintain data integrity and support analytics.

8. Secure SAP & Workday Connectivity

To integrate workforce data from enterprise HR systems, secure connectivity mechanisms are designed. The dashboard communicates with SAP SuccessFactors and Workday using secure APIs and authentication protocols, enabling automatic data synchronization while protecting sensitive employee information.

8.1 SAP SuccessFactors Integration

The dashboard retrieves workforce information from SAP SuccessFactors using REST APIs. OAuth 2.0 is used for authentication, and all communication occurs over HTTPS to ensure secure data transfer. The integration allows employee records, attendance, payroll, leave, and performance data to be synchronized automatically.

Key Features

- REST API communication
- OAuth 2.0 authentication
- HTTPS encryption
- JSON data exchange
- Incremental data synchronization

8.2 SAP Connectivity Workflow

User



Dashboard



Integration Service



OAuth Authentication



SAP SuccessFactors API



Employee Database



Dashboard Reports

The workflow authenticates the application before accessing SAP. Once verified, employee data is retrieved securely and displayed in the analytics dashboard.

8.3 Workday Integration

The system also supports Workday integration using SOAP APIs, REST APIs, and Workday RaaS services. Scheduled synchronization ensures that workforce information remains up to date.

Supported Features

- SOAP API
- REST API
- Workday RaaS
- Scheduled data extraction

8.4 Security Features

The proposed design includes multiple security mechanisms to protect workforce data:

- OAuth 2.0 Authentication
- HTTPS Encryption
- Access Tokens
- Role-Based Access Control (RBAC)
- Secure Credential Storage
- API Logging and Monitoring

These mechanisms ensure that only authorized applications and users can access employee information.

8.5 API Configuration

Parameter	Value
API Type	REST API
Protocol	HTTPS
Data Format	JSON
Authentication	OAuth 2.0
Authorization	Bearer Token

The dashboard sends authenticated API requests and receives employee information in JSON format for further processing.

8.6 Sample API Endpoints

Endpoint	Method	Purpose
/login	POST	User Authentication
/employees	GET	Retrieve Employee Details
/departments	GET	Retrieve Department Information
/attendance	GET	Retrieve Attendance Records
/training	GET	Retrieve Training Information

These endpoints provide secure access to workforce data and support dashboard reporting and analytics.

9. ETL Pipeline Design

The ETL process converts raw workforce data into a structured repository.

Extract



Validate



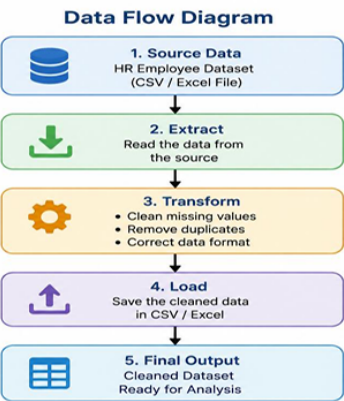
Transform



Load



Analytics



ETL Steps

- Extract data from SAP, Workday, CSV, and Excel.
 - Validate data quality.
 - Standardize data formats.
 - Load data into PostgreSQL.
 - Generate dashboards and reports.
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10. Data Cleansing Workflow

Data cleansing improves the quality of workforce data.

Activities

- Remove duplicate records.
- Handle missing values.
- Validate employee IDs.
- Standardize department names.
- Correct data formats.
- Remove invalid records.

The ETL workflow also includes quality validation before loading cleaned data.

11. Data Transformation Workflow

The transformation process prepares data for analytics.

The data transformation workflow was implemented using Microsoft Power BI (Power Query). The following transformations were performed:

Transformations:

- Imported the IBM HR Analytics dataset into Power BI.
- Verified the data types of all columns.
- Applied Trim to remove unwanted spaces from text fields.
- Applied Clean to remove hidden characters (if performed).
- Created Age Group using a Conditional Column.
- Created Income Level using a Conditional Column based on Monthly Income.
- Validated the transformed dataset.
- Applied Close & Apply to load the transformed dataset for reporting and analysis.

This process ensures consistent, analysis-ready data.

12. Data Storage Workflow

The storage workflow organizes validated data for reporting, dashboards, and AI applications.

The Data Storage Workflow ensures that workforce data collected from various HR systems is stored securely and efficiently in a centralized database for analytics and reporting.

Workflow

HR Data Sources



Raw Data Collection



Data Validation



Data Cleansing



Data Transformation



Validated Dataset



PostgreSQL Database



Analytics Dashboard & AI Models

Description

1. **Data Collection:** Workforce data is collected from SAP, Workday, LMS, payroll systems, and CSV/Excel files.
2. **Data Validation:** Records are checked for completeness, accuracy, and duplicates.
3. **Data Cleansing:** Missing values are handled, duplicate records are removed, and data formats are standardized.
4. **Data Transformation:** Data is converted into a consistent format suitable for analysis.
5. **Data Storage:** The processed data is stored in a centralized PostgreSQL database.
6. **Data Utilization:** The stored data is used for dashboards, reports, machine learning models, and AI-powered workforce analytics.

Storage Layer:

Storage Layer	Purpose
Raw Data Layer	Stores the original workforce data collected from SAP, Workday, CSV files, and other HR data sources without modification.
Cleaned Data Layer	Stores data after removing duplicate records, handling missing values, and correcting data quality issues.
Transformed Data Layer	Stores standardized and transformed data in a consistent format suitable for analysis.
Validated Data Layer	Stores data that has passed all quality validation checks and is ready for loading into the database.
Structured Database Layer	Stores the final processed workforce data in a PostgreSQL database for analytics, dashboards, AI models, and reporting.

Benefits

- Centralized workforce data management
- Improved data quality and consistency
- Secure and scalable storage
- Faster reporting and dashboard generation
- Better support for AI and predictive analytics

13. Expected Deliverables

At the end of Milestone 1, the following deliverables are completed:

- Workforce Data Requirements Document
- System Analytics Specification
- Repository Architecture
- Database Schema and ER Diagram
- ETL & Data Integration Design
- SAP Connectivity Design
- Workday Connectivity Design
- Data Cleansing Workflow
- Data Transformation Workflow

- Data Storage Workflow
 - Final Documentation and Presentation
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14. Conclusion

Milestone 1 establishes the foundational architecture for the AI-Powered Workforce Analytics & Talent Intelligence Dashboard. It defines workforce data requirements, repository architecture, database schema, secure integration mechanisms, ETL workflows, and data quality processes. These deliverables provide a scalable and secure basis for implementing advanced analytics, machine learning models, and AI-powered workforce insights in subsequent milestones.

15. Future Scope

Future milestones will focus on:

- Building ETL pipelines using Python.
- Implementing the PostgreSQL database.
- Developing REST APIs.
- Creating interactive dashboards.
- Training machine learning models for attrition prediction.
- Integrating Retrieval-Augmented Generation (RAG) for conversational analytics.